

**16** (a) Show that the equation  $5x = x^3 - 9$  can be rearranged to give  $x = \sqrt[3]{9 + 5x}$

(1)

- (b) Starting with  $x_0 = 3$ ,  
use the iteration formula  $x_{n+1} = \sqrt[3]{9 + 5x_n}$  three times to find an estimate for a  
solution of  $5x = x^3 - 9$   
Give your answer correct to 3 decimal places.

(3)

(Total for Question 16 is 4 marks)